Module 3 Homework 1 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:15

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Find the value of the next term in this sequence:

$$1, 1, 2, 3, 5, 8, 13, 21, ?$$

- 35
- 28
- 34
- 24
- 33

2

Review Later

Find which of the following statements is true:

- 114×115 ends in 20
- 112×117 ends in 14
- 314×316 ends in 24
- 314×317 ends in 11
- 114×336 ends in 24

3

Review Later

The average of two numbers is a, and the two numbers can be written as $a + c$ and $a - c$. Then, the product of the two numbers is

- $a^2 - c^2$
- a^2

- c^2
- $c^2 - a^2$
- $a^2 + c^2$

4

Review Later

Suppose you have a two-digit number where integer a is the tens digit and integer b is the ones digit, and a and b are both single-digit numbers. Which of the following is an expression for the number that you get from swapping the digits of this number?

- $b + a$
- $10 + a$
- $10 + b$
- $10a + b$
- $10b + a$

5

Review Later

The price of a donut started at $\$1$, and then went up by $x\%$. Later, the price fell by $x\%$. Now the price of the donut is 36% **less** than it was when it started at $\$1$. Find the percent x that it rose and fell by.

- 60%
- 18%
- 72%
- 50%

6

Review Later

Let the operation @ be defined like this:

$$a@b = a - b$$

Find the value of

$$\Big((10@9)@(8@7)\Big)@(6@5)@\Big((4@3)@(2@1)\Big)$$

- 0
- 1
- -1
- -3
- -2

7

Review Later

Let the function

$f(x) = 3x + 6$, if $x < 5$

$f(x) = 7x - 20$, if $x \geq 5$

Find the value of $f(f(f(2)))$.

- 12
- 64
- 428
- 128
- 1024

8

Review Later

How many terms are in the expansion of $2(4x - 4)(4x + 4)$ when it is simplified to lowest terms?

9

Review Later

Find the value of x that makes the following true:

$$8^x \times 9^x = 19^x$$

10

Review Later

Before you take your weekly Daily Challenge exam, which also has 20 questions, your teacher offers to pay you 7 cents for each correct answer you put, minus 3 cents for each incorrect answer you put. In the end, however, she tells you that she owes you zero cents. How many questions did you answer correctly?

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

If $\frac{2}{3}$ of the students in a class take Spanish, $\frac{1}{2}$ of the students in the class take French, and 6 students take both, then if everyone in the class takes at least one of either Spanish or French, how many total students are in the class?

12

Review Later

Which 3-digit integer starting with 6 has this property: that if you delete the first digit, the number formed from the remaining digits is 25 times smaller than the original number?

13

Review Later

Find the value of x if $\frac{13x+11}{x+5} = 7$.

14

Review Later

Alice's brother Bob is twice the age of Alice. Her other brother Charlie is half the age of Alice. In 6 years, Charlie will be half the age of Bob. How old is Alice today?

15

Review Later

Starting from 1000, Tim divides the number that he currently has by 2 and rounds down. Find

the number of times he does this before reaching the number 1.

16

Review Later

Find the largest 2-digit number that has this property: if the last digit is deleted, then the number is divisible by the new number.

17

Review Later

A rich merchant has many gold coins and doesn't want anybody to know about them. One day his wife asked, "How many gold coins do you have?" The merchant replied, "If I divide the coins into two unequal numbers, then 36 times the difference between the two numbers equals the difference between the squares of the two numbers." How many gold coins does the merchant have?

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

A dishwasher that costs $\$3600$ normally is on sale for 35% off. An oven that normally costs $\$10,400$ is now on sale for 15% off. If I buy both the dishwasher and the oven at their discounted prices, what fraction of the normal price total am I paying? Your answer should be a fraction without spaces, e.g. 1/2

19

Review Later

Two numbers are in the ratio $1:3$. The first number starts with 1, and the second number is produced by altering the first number in this way: the 1 at the start is removed and placed at the end. What is the smallest possible value of the smaller number, which starts with 1?

20

Review Later

Find the value of

$$\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+\frac{5}{2^5}+\ldots$$

View Marked Questions

Finished

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